

TITLE:

Interactive Plotly/Dash dashboard

Dataset Used:

nhstalkingtherapies_month_mar_2026_activity_performance.csv

&

psych-ther-ann-rep-csv-2024-25-psych-meds-2

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Interactive Plotly/Dash Dashboard

Introduction

IAPT Dataset Introduction

The IAPT dataset contains information related to NHS Talking Therapies services across England. It includes:

Main Information

- Referrals received
- Treatment activity
- Recovery rates
- Waiting times
- Completed treatments
- Regional service performance

Prescribing Dataset Introduction

The prescribing dataset contains data about antidepressant and anxiolytic medication prescribing trends.

Main Information

- Prescription counts
- Medication categories
- Regional prescribing activity
- Monthly prescribing trends

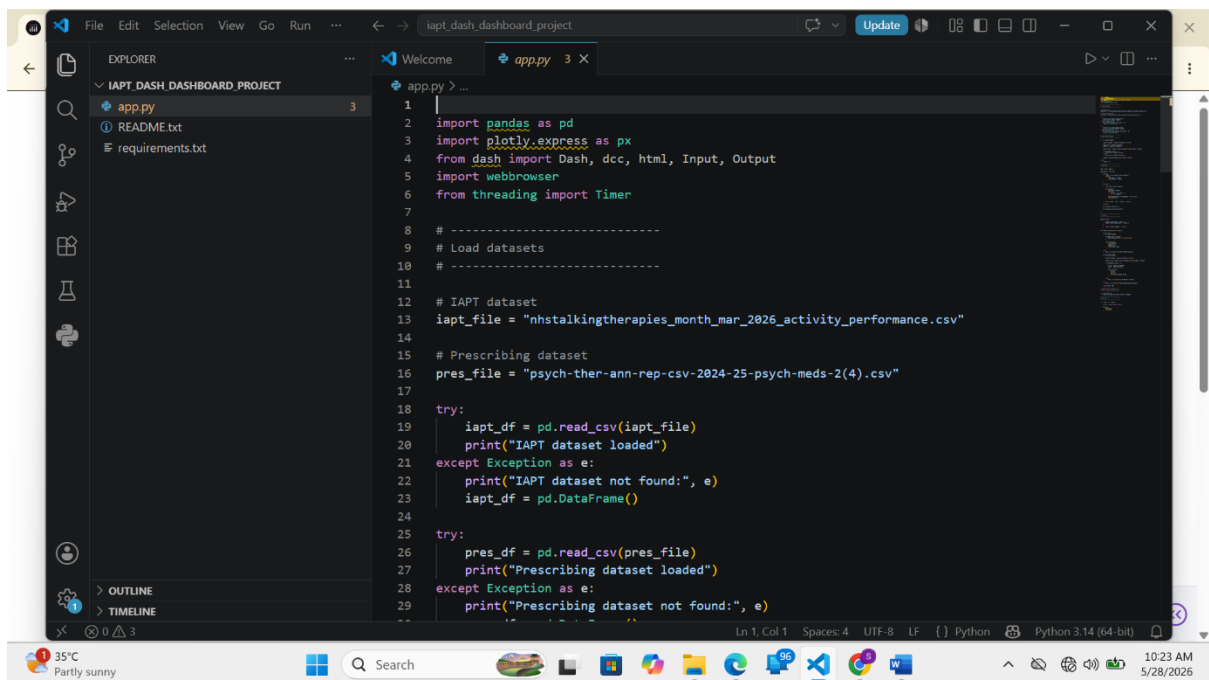
Technologies Used

Technology

Python	-	Core programming language
Pandas	-	Data cleaning and preprocessing
Plotly	-	Interactive charts and visualizations
Dash	-	Web dashboard framework

Jupyter/VS Code	-	Development environment
NumPy	-	Numerical operations
HTML Components (Dash)	-	Dashboard layout
CSS Styling (Optional)	-	UI enhancement

Interactive Plotly/Dash Dashboard



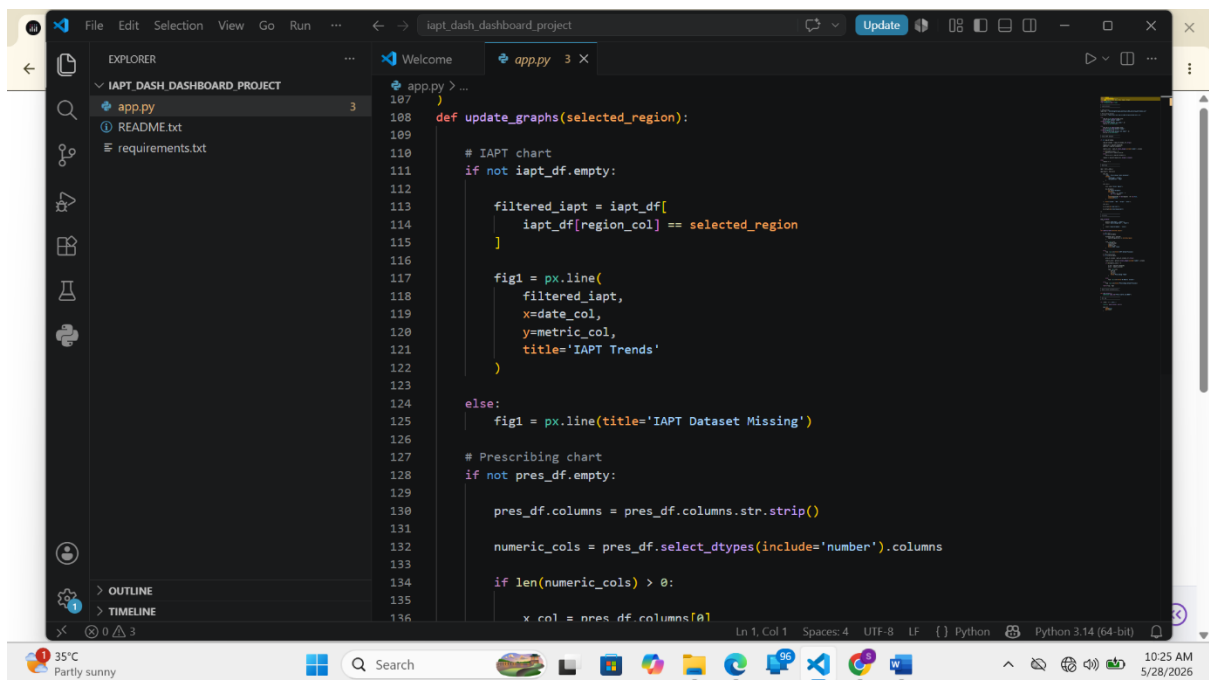
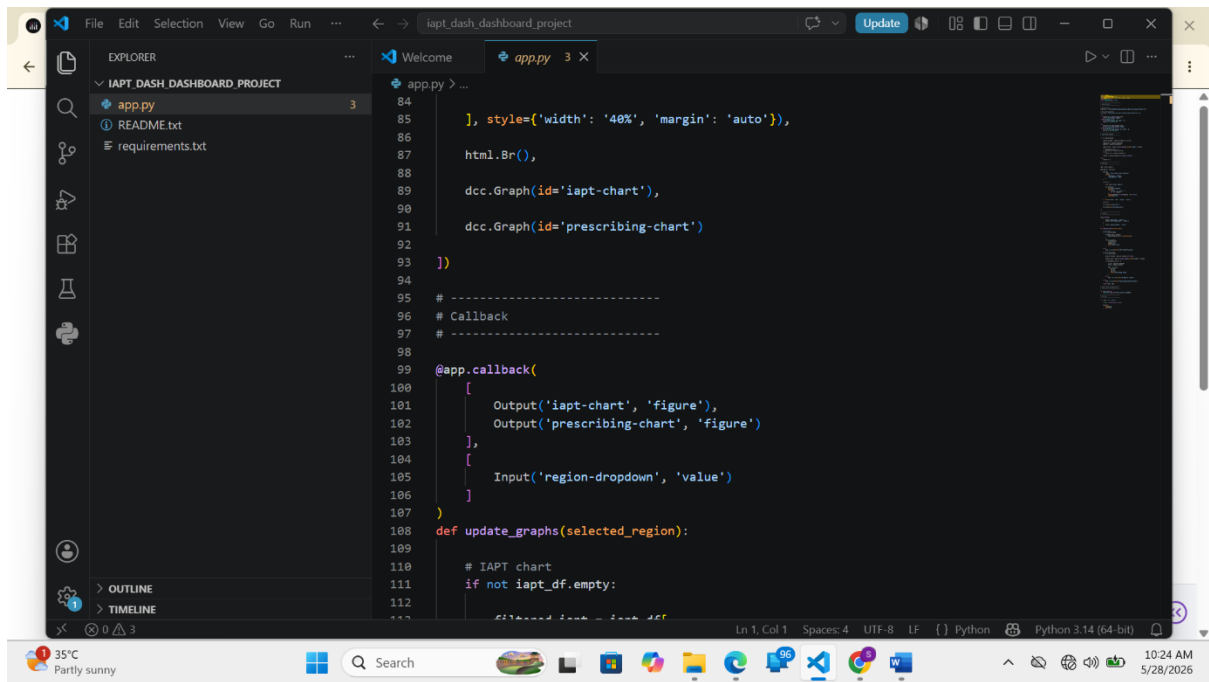
```

1
2 import pandas as pd
3 import plotly.express as px
4 from dash import Dash, dcc, html, Input, Output
5 import webbrowser
6 from threading import Timer
7
8 # -----
9 # Load datasets
10 # -----
11
12 # IAPT dataset
13 iapt_file = "nhstalkingtherapies_month_mar_2026_activity_performance.csv"
14
15 # Prescribing dataset
16 pres_file = "psych-ther-ann-rep-csv-2024-25-psych-meds-2(4).csv"
17
18 try:
19     iapt_df = pd.read_csv(iapt_file)
20     print("IAPT dataset loaded")
21 except Exception as e:
22     print("IAPT dataset not found:", e)
23     iapt_df = pd.DataFrame()
24
25 try:
26     pres_df = pd.read_csv(pres_file)
27     print("Prescribing dataset loaded")
28 except Exception as e:
29     print("Prescribing dataset not found:", e)

```

```
29 print("Prescribing dataset not found:", e)
30 pres_df = pd.DataFrame()
31
32 # -----
33 # Clean IAPT dataset
34 # -----
35
36 if not iapt_df.empty:
37
38     iapt_df.columns = iapt_df.columns.str.strip()
39
40     region_col = iapt_df.columns[0]
41     date_col = iapt_df.columns[1]
42
43     numeric_cols = iapt_df.select_dtypes(include='number').columns
44
45     if len(numeric_cols) > 0:
46         metric_col = numeric_cols[0]
47     else:
48         metric_col = iapt_df.columns[-1]
49
50     regions = iapt_df[region_col].dropna().unique()
51
52 else:
53     regions = []
54
55 # -----
56 # Dash App
57 # -----
```

```
56 # Dash App
57 # -----
58
59 app = Dash(__name__)
60
61 app.layout = html.Div([
62
63     html.H1(
64         "IAPT + Prescribing Trends Dashboard",
65         style={
66             "textAlign": "center",
67             "marginBottom": "30px"
68         }
69     ),
70
71     html.Div([
72
73         html.Label("Select Region"),
74
75         dcc.Dropdown(
76             id='region-dropdown',
77             options=[
78                 {'label': r, 'value': r}
79                 for r in regions
80             ],
81             value=regions[0] if len(regions) > 0 else None,
82             clearable=False
83         )
84     ])
85 ])
```



This screenshot shows the Visual Studio Code editor with the file `app.py` open. The Explorer sidebar on the left shows the project structure for `IAPT_DASH_DASHBOARD_PROJECT`, including `README.txt` and `requirements.txt`. The main editor window displays the `update_graphs(selected_region)` function. The function logic is as follows:

```
def update_graphs(selected_region):
    x_col = pres_df.columns[0]
    y_col = numeric_cols[0]

    fig2 = px.line(
        pres_df,
        x=x_col,
        y=y_col,
        title='Prescribing Trends'
    )

    else:
        fig2 = px.line(title='No Numeric Columns')

    else:
        fig2 = px.line(title='Prescribing Dataset Missing')

    return fig1, fig2

# -----
# Open browser automatically
# -----
def open_browser():
    webbrowser.open_new("http://127.0.0.1:8050/")

# -----
# Run app
# -----
```

The status bar at the bottom indicates the current position is Line 1, Column 1, with 4 spaces, in UTF-8 encoding, using the Python 3.14 (64-bit) interpreter.

This screenshot shows the same Visual Studio Code editor with `app.py` open, displaying the code from line 158 to 173. The `open_browser` function and the main execution block are visible:

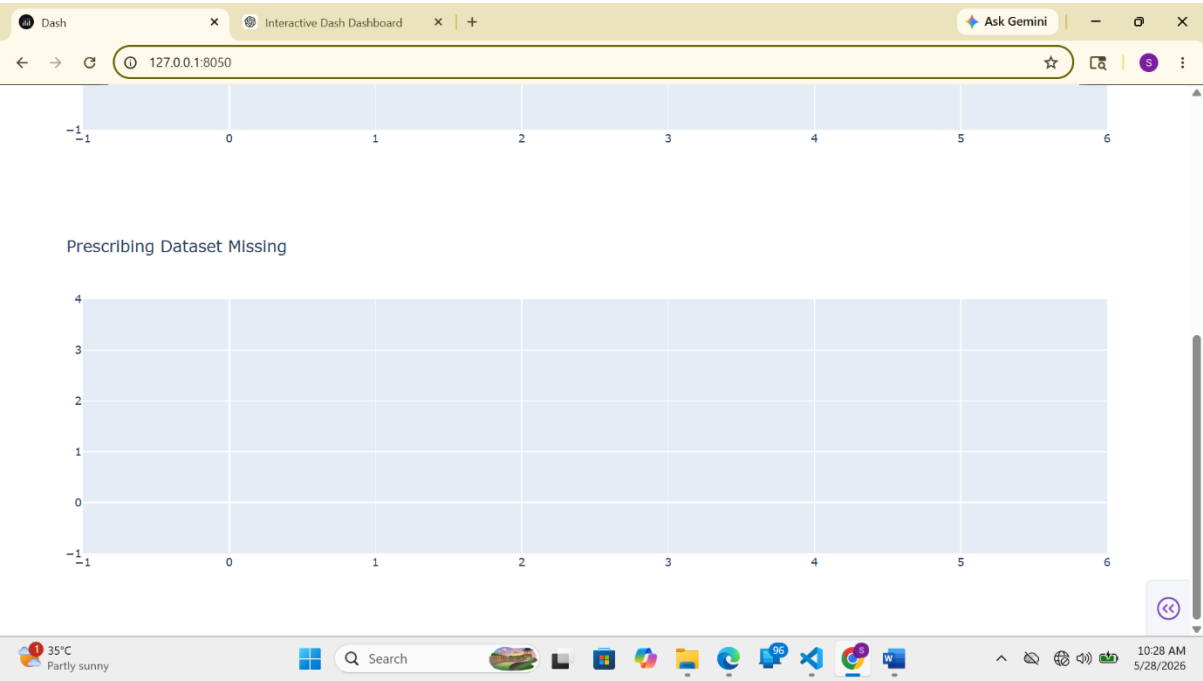
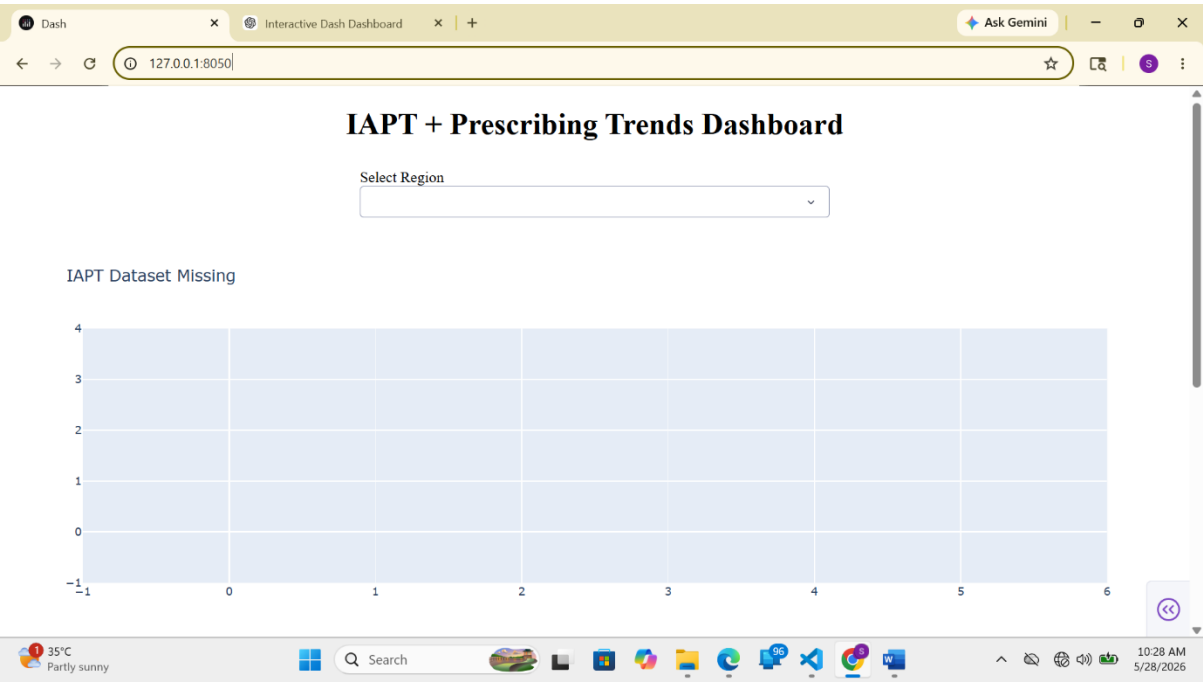
```
def open_browser():
    webbrowser.open_new("http://127.0.0.1:8050/")

# -----
# Run app
# -----
if __name__ == "__main__":
    Timer(1, open_browser).start()

    app.run(
        debug=True,
        port=8050
    )
```

The status bar at the bottom shows the same configuration as the first screenshot.

Output



Conclusion

The Interactive Plotly Dash Dashboard successfully combines IAPT and prescribing datasets into a single analytical platform. Using Python, Pandas, Plotly, and Dash, the project provides an efficient way to explore mental health service trends and prescribing patterns.

The dashboard improves data visualization and user interaction through dropdown filters and dynamic charts. It enables users to monitor healthcare performance, identify regional differences, and analyze prescribing behavior effectively.

This project demonstrates how modern data visualization tools can transform raw healthcare datasets into meaningful insights for decision-making and reporting.